

UID EXPERIMENT-2

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CLI

```
tasks = []

def add_task(task):
    tasks.append(task)
    print(f"Task '{task}' added.")

def view_tasks():
    if tasks:
        print("Your tasks:")
        for idx, task in enumerate(tasks, 1):
            print(f"{idx}. {task}")
    else:
        print("No tasks to show.")

def remove_task(task_number):
    if 0 < task_number <= len(tasks):
        removed_task = tasks.pop(task_number - 1)
        print(f"Task '{removed_task}' removed.")
    else:
        print("Invalid task number.")

def main():
    while True:
        print("\nOptions:")
        print("1. Add Task")
        print("2. View Tasks")
        print("3. Remove Task")
        print("4. Exit")

        choice = input("Enter your choice: ")

        if choice == '1':
            task = input("Enter task: ")
            add_task(task)

        elif choice == '2':
            view_tasks()

        elif choice == '3':
            task_number = int(input("Enter task number to remove: "))
            remove_task(task_number)

        elif choice == '4':
            print("Exiting...")
            break
```

```
        else:
            print("Invalid choice. Please try again.")

if __name__ == "__main__":
    main()
```

OUTPUT:

```
Options:
1. Add Task
2. View Tasks
3. Remove Task
4. Exit
Enter your choice: 1
Enter task: read
Task 'read' added.
```

```
Options:
1. Add Task
2. View Tasks
3. Remove Task
4. Exit
Enter your choice: 2
Your tasks:
1. read
```

```
Options:
1. Add Task
2. View Tasks
3. Remove Task
4. Exit
Enter your choice: 1
Enter task: write
Task 'write' added.
```

```
Options:
1. Add Task
2. View Tasks
3. Remove Task
4. Exit
Enter your choice: 3
Enter task number to remove: 1
Task 'read' removed.
```

```
Options:
1. Add Task
2. View Tasks
3. Remove Task
4. Exit
Enter your choice: 4
Exiting...
|
```

GUI

```
import tkinter as tk
from tkinter import messagebox

tasks = []

def add_task():
    task = task_entry.get()
    if task:
        tasks.append(task)
        task_entry.delete(0, tk.END)
        update_task_list()
    else:
        messagebox.showwarning("Warning", "Task cannot be empty")

def update_task_list():
    task_list.delete(0, tk.END)
    for task in tasks:
        task_list.insert(tk.END, task)

def remove_task():
    selected_task_index = task_list.curselection()
    if selected_task_index:
        tasks.pop(selected_task_index[0])
        update_task_list()
    else:
        messagebox.showwarning("Warning", "Please select a task to remove")

# Create window
app = tk.Tk()
app.title("To-Do List")
app.geometry("350x300")

# Task entry box
task_entry = tk.Entry(app, width=40)
task_entry.pack(pady=10)

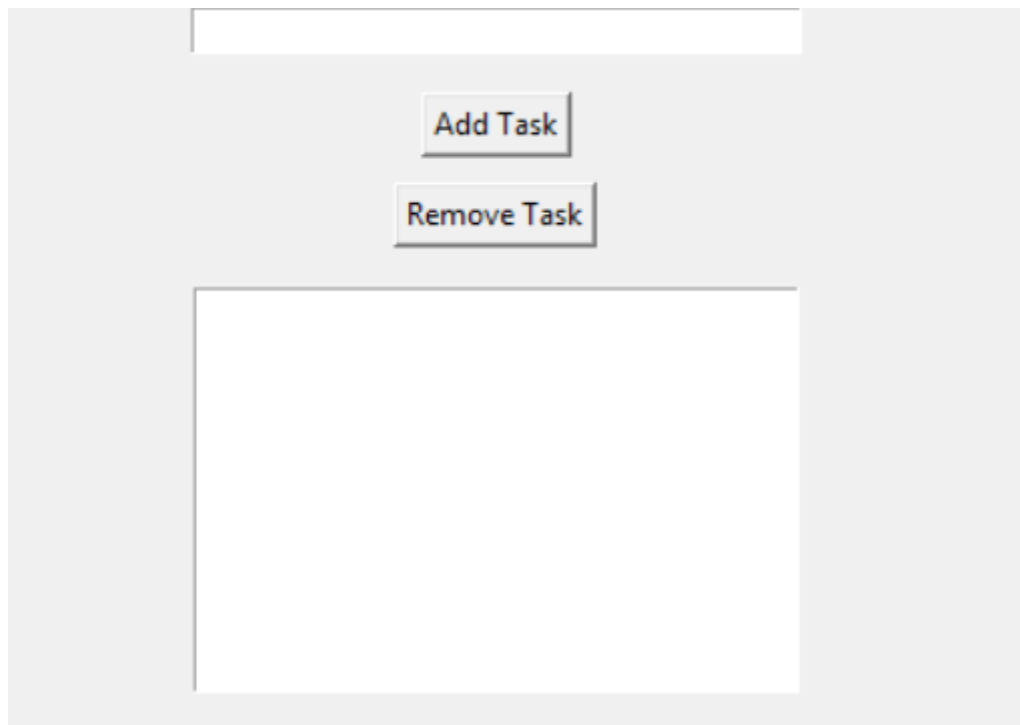
# Add button
add_button = tk.Button(app, text="Add Task", command=add_task)
add_button.pack(pady=5)

# Remove button
remove_button = tk.Button(app, text="Remove Task", command=remove_task)
remove_button.pack(pady=5)

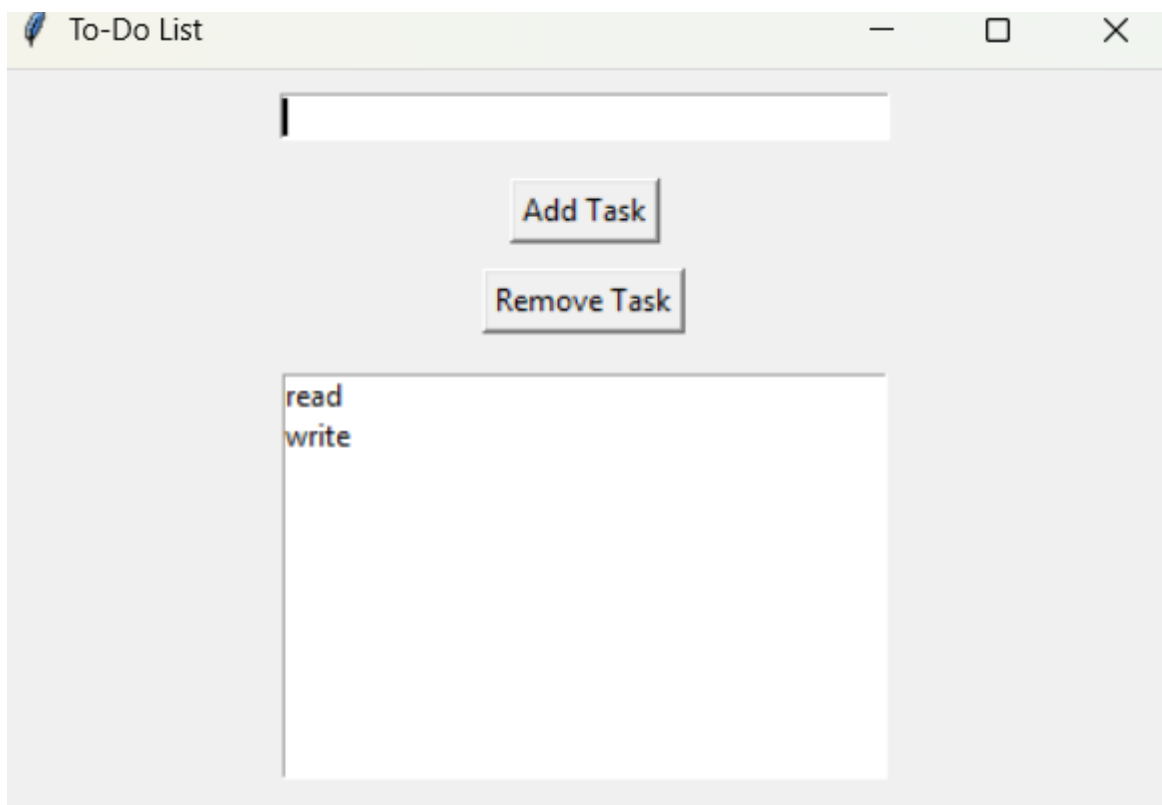
# Task list display
task_list = tk.Listbox(app, width=40, height=10)
task_list.pack(pady=10)

# Run application
app.mainloop()
```

OUTPUT:



A simple web form with a light gray background. At the top is a white text input field. Below it are two buttons: "Add Task" and "Remove Task". At the bottom is a large, empty white rectangular area for displaying tasks.



A screenshot of a web application window titled "To-Do List". The window has a standard title bar with minimize, maximize, and close buttons. The form inside is identical to the one above, but the large white area now contains the text "read" and "write" stacked vertically.



To-Do List



Add Task

Remove Task

read

VUI

```
import speech_recognition as sr
import pytttsx3
import sounddevice as sd
import numpy as np

tasks = []

engine = pytttsx3.init()
recognizer = sr.Recognizer()

def speak(text):
    engine.say(text)
    engine.runAndWait()

def record_audio():
    samplerate = 16000
    duration = 5
    print("Listening...")
    audio = sd.rec(int(duration * samplerate), samplerate=samplerate, channels=1)
    sd.wait()
    return audio

def recognize_speech():
    try:
        audio_data = record_audio()
        audio_bytes = audio_data.tobytes()
        audio = sr.AudioData(audio_bytes, 16000, 2)
        text = recognizer.recognize_google(audio)
        print("You said:", text)
        return text.lower()
    except:
        speak("Sorry I could not understand")
        return None

def add_task(task):
    tasks.append(task)
    speak(f"Task {task} added")

def view_tasks():
    if tasks:
        speak("Your tasks are")
        for t in tasks:
            speak(t)
    else:
        speak("No tasks available")
```

```

def remove_task(n):
    if 0 < n <= len(tasks):
        removed = tasks.pop(n-1)
        speak(f"Task {removed} removed")
    else:
        speak("Invalid task number")

def main():
    while True:
        speak("Say add task, view tasks, remove task or exit")
        command = recognize_speech()

        if not command:
            continue

        if "add task" in command:
            speak("What task")
            task = recognize_speech()
            if task:
                add_task(task)

        elif "view task" in command:
            view_tasks()

        elif "remove task" in command:
            speak("Say task number")
            num = recognize_speech()
            if num and num.isdigit():
                remove_task(int(num))

        elif "exit" in command:
            speak("Exiting program")
            break

        else:
            speak("Command not recognized")

if __name__ == "__main__":
    main()

```

OUTPUT:

```
Listening...
You said: add task
Listening...
You said: read
Listening...
You said: add task
Listening...
You said: right
Listening...
You said: voice
Listening...
Listening...
Listening...
Listening...
You said: new task
Listening...
Listening...
Listening...
Listening...
Listening...
You said: new task
Listening...
You said: exit
```